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Virtual Reality Induced Symptoms and Effects (VRISE) [11]. These are related but not identical, differing mainly in their causes. Because our work addresses the broad issue of sickness experienced during VR immersion, without restricting its specific origin, we use the general term *VR sickness* throughout this paper [24].

Although extensive research has explored the origins of VR sickness and developed numerous techniques to mitigate it [5], far fewer innovations have emerged in how it is measured. Existing approaches can be broadly classified into objective and subjective methods [9]. Objective methods have lately expanded due to recent advances in biosensors (e.g., Galvanic Skin Response, heart rate, eye tracking), yet these signals are costly, technically demanding, and difficult to interpret in real time. Conversely, subjective assessments remain dominated by questionnaires and scales. These address the three dimensions of VR sickness into which symptoms are typically clustered: nausea (for symptoms related to stomach awareness, increased salivation, burping), disorientation (for symptoms related to dizziness, vertigo), and oculomotor (for symptoms related to eyestrain, difficulty focusing, blurred vision, headache) [14, 25, 29]. Questionnaires and scales are usually applied before and after the VR experience, thus failing to capture the temporal evolution of the symptoms. Alternatively, they may also be applied halfway through, requiring the users to take out their head-mounted displays (HMD) thereby disrupting immersion. Real-time information of VR sickness could support not only retrospective evaluation of experience designs, but also the development of user-centered adaptive immersive systems in which scenes respond dynamically to the users' current symptoms. This gap highlights the need for immersive self-report tools capable of monitoring symptoms continuously throughout the VR session.

Recent research has proposed embedding questionnaires or scales directly inside the VR environment [29, 36, 40], developing tools that facilitate less intrusive and more frequent data collection [26, 31, 41, 49]. However, these approaches either still rely on discrete measures, or fail to capture all three dimensions of VR sickness and therefore, do not provide a comprehensive method for continuous self-report of VR sickness during immersion.

Continuous self-reporting, on the other hand, offers the possibility of a more intuitive and cognitively less demanding interaction with a graphical interface [47], and has been explored in other areas of Human-Computer Interaction (HCI). Existing methods can be classified into four types: **"Linear models"**, which encode intensity along a single axis, typically using a bar-like representation [4, 13, 16, 19, 49]; **"Circle models"**, where the diameter of a circle or halo reflects the magnitude of the signal [50–53, 55]; **"Rotary models"** which feature rotary or angular indicators reminiscent of kitchen knobs or automotive speedometers, and are often operated via dial style controllers [1, 10, 30]; and **"Quadrant-based models"**, which map data onto a two-dimensional Cartesian plane, typically representing arousal and valence [2, 12, 18, 20, 32, 43–46, 48, 54].

In this work, we investigate the question of how to design a continuous interface for real-time self-reporting of VR sickness, comparable with current validated questionnaires but without disrupting immersion. Inspired by the approaches of other areas in HCI and following a user-centered design methodology [17, 34, 42], we developed three design principles (see Section 2) that served as heuristics to narrow down the design space and guide us to design three different interface prototypes. This allows us to validate the feasibility of continuous VR sickness self-reporting, while also identifying which of the three designs works best according to user-centered evaluation criteria. Our findings show that users significantly prefer the fluid nature of our continuous real-time reporting over discrete alternatives, finding it more intuitive and less cognitively demanding, while also reducing disruption.

2 Design Principles

We take into account three main design principles, which will guide the placement, behavior, and feedback logic of our interface.

P1. Design for a good VR experience. Rebelo et al. [38] defined the *i3 VR* user-experience triangle: immersion, interaction, and imagination. Regarding *immersion*, and since our interface will be used inside the VR environment, it must remain visible without becoming distracting. To support *interaction*, all its elements must offer a comfortable, predictable, and consistent behavior, maintaining a unified logic to reduce cognitive effort and prevent errors. Last, to support *imagination*, the interface should add as little mental workload as possible, so that users can direct their attentional resources and engagement towards the virtual environment itself.

P2. Design for Universal Access and Human-Centered Interaction guidelines. Our interface should follow established Human-Centered Interaction guidelines, for which we turn to Jerald's book on Human-Centered Design for Virtual Reality [24]. It should also follow Norman's foundational usability principles [33]. Together, these guidelines allow us to enable universal access, comfortable even under varying levels of VR sickness. The specific visual, functional, and interaction decisions informed by these principles are detailed in the following sections.

P3. Design for a ground truth. Lastly, the effectiveness of our interface must be comparable to well-established VR sickness questionnaires, without breaking immersion. The Simulator Sickness Questionnaire (SSQ) [25] has been widely used in VR research. However, it was developed as an adaptation of the earlier Motion Sickness Questionnaire (MSQ) [37], and does not entirely match the phenomenology of VR-induced symptoms [39]. In response to these limitations, instead, the shorter Cybersickness in VR Questionnaire (CSQ-VR) (based on 7-point Likert Scale) was explicitly developed and validated within immersive VR contexts [15, 28, 29]. We thus select CSQ-VR as ground-truth questionnaire to evaluate our proposal.

Conclusion. Based on these principles, we narrow down the design space of our interface by focusing on three of the four types identified in the Introduction: Linear, Circle, and Rotary models. We discard the fourth option, Quadrant-based models, since VR sickness does not map meaningfully onto two-axis spaces.

3 Interface Adaptations

We make the following adaptations to the three remaining types of interfaces, to accommodate for the particular case of VR sickness.

Visual design. We design three interfaces, inspired by existing continuous self-reporting techniques: *Bars* (derived from Linear models and implemented as vertical bars), *Halos* (based on the Circle family), and *Dials* (a semicircular gauge implemented as an adaptation of the Rotary model). Each prototype consists of three

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distinct elements, allowing users to independently report the three symptom dimensions of VR sickness: nausea, disorientation, and oculomotor. These interfaces are shown in Fig. 1(a).

Semantic layer. By default, no textual labels are displayed; instead, we assign a distinct icon to convey each symptom, minimizing visual load and supporting rapid recognition [1, 6, 7, 46, 48, 50]. When needed, users can activate an on-demand wording reference by pressing a button, temporarily showing a brief label describing each of the three symptom dimensions. We use Atkinsons Hyperlegible, a typeface recommended by the Braille Institute for visually impaired users [35].

Visual encoding. We assign a discriminable color to each symptom [22], chosen to remain distinguishable even under color-blindness conditions (HEX values: yellow-green #A2C523 for nausea; soft blue #3FA7D6 for disorientation; violet #6A1B9A for oculomotor). We use low-saturation tones, subtle shadows, and controlled transparency to reduce visual fatigue and support divided attention, allowing the interface to remain perceptible without competing with the virtual scene [21].

Functional design. All interface elements are head locked within the ergonomic field of view, positioned in a lower centered location for constant visibility. Interaction is performed via the HMD controller: users point at a symptom element, select it using the trigger, and adjust the intensity using the joystick (chosen for its return spring mechanisms, which allow users to sense the neutral position without visual confirmation, again reducing cognitive load [2, 39]). Interface feedback includes audiovisual cues: a brief sound plus white highlight when hovering, red highlight during selection via the trigger.

4 Evaluating the design

Prior to the formal study, we conducted several feedback pilot rounds with three HCI researchers. The formal study was reviewed by our institute's regional ethics committee. The within-subjects design involved three sessions on separate days, one for each interface design in random order, to mitigate potential memory issues. Before participation, all individuals reviewed and signed the informed consent form and received instructions about the study's objectives and procedure. The study was divided in three phases: Questionnaires; VR experience; and Feedback. In the first phase, participants completed an online questionnaire collecting demographics, their past experience with immersive environments, as well as the Visually Induced Motion Sickness Susceptibility Questionnaire (VIMSSQ) [27] to assess their predisposition to sickness and their current state. Next, participants sat in a non-rotating chair with armrests, since studies indicate that standing induces more VR sickness compared to a seated posture [3, 57], and received instructions on how to put on and calibrate the Varjo XR-4 HMD and controller. At the beginning of the second phase, participants spent the first five minutes in a training scene, designed to help them familiarize with VR, learn how to interact with the interface, and understand the three VR symptom dimensions and their mapping to the on-demand wording reference (see supplementary material S.1 for a description of the training scene). After that, the actual test begins. Participants were asked to annotate their symptoms using the given interface while immersed in the Cybersicker amusement ride scene (see Fig. 1 (b)), which was selected because it was specifically designed to study induced effects of VR sickness [56]. This lasted ten minutes, since cybersickness symptoms are typically noticed within this timeframe [11, 23]. We modified the motion speed and acceleration of the scene to elicit varying levels of discomfort and capture dynamic symptom fluctuations (see S.2 of the supplementary material for a description of the speed modification in the scene). All scene parameters, together with the user's current symptoms intensities, are continuously logged to enable the subsequent evaluation of the three interfaces designs. To minimize external factors, no music was played during this phase [28]. Participants verbally answered the CSQ-VR at the beginning, middle and at the end of the experience [49], but without removing the HMD or introducing additional on-screen interaction. They could stop the experiment at any time.

Once the experience was over, the HMD and controller were taken off, and participants completed the third phase by answering the System Usability Scale (SUS) [8] for both the interface used and the 7-point Likert Scale CSQ-VR, followed by a brief, semi-structured interview to collect qualitative feedback and identify potential interface improvements. In total, the duration was approximately 45 minutes on day one, and 25 minutes on days two and three, since less training was required. After testing the three interfaces, participants ranked them during the interview.

4.1 Results

We evaluate the proposed interfaces according to three main objectives:

O1. Ground truth agreement: Assessing whether continuous interfaces are comparable to the CSQ-VR questionnaire used as ground truth. **O2. Temporal sensitivity:** Examining if the proposed interfaces capture the continuous evolution of symptoms better than discrete traditional questionnaires. **O3. User experience comparison:** Identifying which of the three designs (Bars, Halos or Dials) achieves better usability, acceptance, and comfort.

A total of 25 participants (average age = 27.12, M = 15, F = 10) took part. Of these, 56% reported frequent prior use of VR, 40% reported limited prior experience (less than 5 uses), and 4% reported no prior VR exposure. One participant had cochlear implants due to hearing impairment, and two participants reported color-blindness. We analyze data collected from the CSQ-VR questionnaire, the continuous interface, and the interviews to evaluate the proposed designs.

O1. Ground truth agreement. To evaluate the accuracy of VR sickness estimates provided by our interfaces, we computed the mean absolute error (MAE) between interface-based measurements and the scores obtained from the standardized CSQ-VR, administered at three discrete time points. Fig. 2 (left) summarizes MAE results across the three interfaces (MAE values range from 0 to 1, with lower values indicating closer agreement with the reference measure). Normality assumptions were mildly violated (Shapiro–Wilk test on within-subject residuals, $p = .024$); therefore, we report non-parametric analyses. A Friedman test ($\chi^2(2) = 3.44$, $p = .18$) did not reveal significant differences in MAE across interface conditions. Post-hoc pairwise comparisons confirmed the absence of meaningful differences between interfaces. Overall, all interface conditions

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